Target-Driven Iterative Autoencoder for Hyperspectral Target Detection

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***Abstract*— With the advantages of being scalable, labeled-sample free, and structurally simple, autoencoder (AE) is suitable for hyperspectral target detection (HTD) with discriminative attributes extraction ability. However, there are some bottlenecks in current AE-based approaches, like network complexity, parameter determination, and dictionary reliability problems. In this article, a novel target-driven AE (TAE) with a constrained latent code layer and improved loss function is proposed. To achieve effective noise and interference annihilation, a simple AE-based network is designed with the adaptively determined number of hidden neurons by estimating the ranks of the background (BKG) and the target component. In addition, to untangle the dependence on the BKG dictionary, a novel strategy is imposed on the traditional Kullback–Leibler (KL) divergence, called the truncated KL divergence (TKL), is designed and imposed by the loss function of TAE, guiding the propensity for network reconstruction of the targets over BKG with probabilities rather than dictionaries. Furthermore, unlike AE-based methods with forward strategy, a novel target-driven iterative AE (ITAE) framework is developed to further exploit the potential of prior and posterior knowledge. The iterative system repeatedly trains the same TAE and adds the weighted constrained energy minimization (CEM) detection map back to the current data matrix for the next iteration. ITAE provides a general classic-deep learning collaborative framework that can train any AE with a traditional detector through an iterative process to improve the diversity between the targets and the BKG. To validate the effectiveness of ITAE, six other state-of-the-art HTD methods are chosen for comparison under six datasets of various scenarios. Based on the 3-D receiver operating characteristic (ROC) curve-derived evaluation metric, we conducted experiments for parameter analysis, detection performance comparison, noisy study, and ablation study. Results show that ITAE outperforms other methods, especially in target detectability.**

***Index Terms*— Autoencoder (AE), constrained energy minimization (CEM), hyperspectral target detection (HTD), iterative AE.**

NOMENCLATURE

AE	Autoencoder.
AUC	Area under ROC curve.
BKG	Background.
BS	Background suppressibility.
CEM	Constrained energy minimization.
HFC	Harsanyi–Farrand–Chang.
HSI	Hyperspectral image.
HTD	Hyperspectral target detection.
ITAE	Target-driven iterative AE.
ITAE-1	First iteration in ITAE.
ITAE-i	ith iteration in ITAE.
ITAE-N	Last iteration in ITAE.
$\mathbf{M}^{(i)}$	Detection map at the ith iteration in ITAE.
NPD	Neyman–Pearson detector.
NWHFC	Noise-whitened HFC.
ROC	Receiver operating characteristic.
SDIN	Signal decomposition interference noise.
TAE	Target-driven autoencoder.
TD	Target detectability.
TKL	Truncated KL divergence.
TI	Tanimoto index.
$\mathrm{TI}^{(i)}$	TI value at the ith iteration in ITAE.
VD	Virtual dimensionality.
$\mathbf{X}^{(i)}$	Input matrix at the ith iteration in ITAE.
$\hat{\mathbf{X}}$	Reconstructed data of $\mathbf{X}$.

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I. INTRODUCTION

HYPERSPECTRAL imaging provides abundant spatial and spectral information, spanning from visible to infrared bands [1]. Thus, it is a valuable tool for revealing essential characteristics of materials that are not visible to visual inspection. This capability can be used for various tasks, such as HTD [2], hyperspectral anomaly detection (HAD) [3], hyperspectral change detection (HCD) [3], and hyperspectral image classification (HSIC) [4]. Specifically, HTD focuses on detecting if pixels are targets of interest or noninterest based on preknown knowledge.

In recent decades, a multitude of HTD methods have been developed to enhance detection accuracy and robustness, with some pioneering advancements in various directions based on different hypotheses and theories. For instance, the CEM method proposed by Harsanyi [5] utilizes the second-order statistics of HSIs to suppress all nontarget signals, representing and eliminating the BKG by inverting the correlation matrix. Due to its excellent generalization ability and relatively weak discriminative ability, CEM often serves as a benchmark for improvement. The key problem here is that in complex

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scenarios, nontarget signals cannot be simply represented as second-order statistics, and the noise and interference components can introduce misleading information into the detection process. To address this issue, assorted methods were developed based on matrix decomposition [6], component representation [7], and contemporary machine learning (ML) and deep learning (DL) theories [8]. These enhanced strategies map or decompose the original HSIs into a new space to improve the discrimination of targets, facilitating differentiation. The specific related works will be discussed more concretely in Section II.

Among those methods, those based on ML and DL have gained significant attention due to their exceptional capabilities in generalization, feature extraction, and representing nonlinear correlations. These methods employ advanced techniques to extract spectral, spatial, and contextual features from HSIs, which are then used to classify target and nontarget pixels. However, ML- or DL-based models typically require a substantial amount of labeled samples to achieve satisfactory performance, which is often challenging to obtain in the context of HSIs. Consequently, unsupervised algorithms like AE are more suitable for HTD scenarios with scarce labeled data. As a structurally simple network, AE does not require labeled data to extract discriminative features, offering the advantages of high scalability. Leveraging by the physical characteristics of HSIs, AE can effectively extract and infer the signals of interest, thereby producing higher quality images. Other than HTD, AE has been widely used in varying HSI processing tasks like unmixing [9] and anomaly detection (AD) [10], [11].

Currently, there are two primary objectives for AE improvements in HSI processing tasks. The first is to eliminate the noise and interference components in HSIs, and the second is to enhance the diversity between the targets and the BKG, both crucial for effective detection. Various strategies have been adopted to upgrade AE, including optimizing network structures. For instance, the hierarchical denoising AE (HDAE) [12] introduces additional dropout layers to create noisy vectors hierarchically, encouraging the network to explore deeper noise features for enhanced denoising capability. And the multiscale denoising AE (MSDAE) [13] combines a multiscale approach with DAE to leverage invariant information in target detection. However, these approaches often necessitate deep, multilayer structures to achieve satisfactory noise reduction, resulting in increased network complexity and challenges in parameter determination. To suppress BKG and enhance targets, method with macro-microfeature extracted by 3-D residual AE (3-D-MMRAE) in [14] segments the HSI into patches of different sizes, feeding them into two separate brunches for multiscale feature extraction, which helps to identify the target from nontarget pixels. Apart from the computation complexity challenge, the main obstacle this method faces is the determination of crucial parameters like patch sizes and feature dimensionality, which limits its generalization ability severely.

In addition, another direction for AE improvement is to modify the loss function to constrain and guide the network, defining the reconstruction priority to obtain more discriminative targets while suppressing other nontarget signals. For example, the discriminative feature learning with distance constrained stacked sparse AE (DCSSAE) in [15] first utilizes the endmember extraction method to develop BKG and target dictionaries, which helps to measure the distance between training samples and targets in the improved loss function. With a high dependence on the reliable dictionary construction, the method performance will descend severely when the extraction of endmembers is inaccurate and inadequate.

To address the above issues and better exploit AE in HTD tasks, a structurally simple network with no dependence on the dictionary construction and adaptive parameters, called the TAE, is proposed. With a constrained latent code layer and improved loss function, TAE can eliminate noise and interference and smooth the BKG adaptively. We consider AE as a process of data compression and reconstruction from the perspective of compressed sensing. It shows that the properties of the latent features of AE directly affect the effectiveness and tendency of data reconstruction. Supported by the component analysis theory, we propose to obtain an effective representation and reconstruction of targets of interest and BKG for detector improvement. Inspired by the distinct signal source estimation method in [16], the hidden dimension of TAE is adaptively determined by the estimated VD, which represents the rank of BKG and target components, and the detailed description is shown in Section III-A1. In addition, inspired by the HAD method of guided AE proposed by Xiang et al. [17], we want to drive the network to infer details of the targets, rather than the BKG. For this purpose, we designed an improved loss function for TAE, where the degree of influence of each pixel on the overall reconstruction is jointly determined by prior and posterior knowledge as a weighting parameter. In this process, we propose a novel strategy imposed on the traditional Kullback–Leibler (KL) divergence, called the TKL, whose distribution is optimized by analyzing the statistical characteristic. The optimization of the loss function is designed to steer the TAE toward prioritizing target reconstruction while de-emphasizing BKG details, thereby facilitating BKG smoothing through reconstruction.

Furthermore, current works of AE in HTD are all unidirectional, which limits the utilization of prior and posterior knowledge. To break the bottleneck of detection performance, we investigate the untapped potential of structural diagrams in recent literature, like the ensemble framework utilized in [18], [19], and [20], the hierarchical framework listed in [12], [21], [22], and [23], and the iterative framework listed in [24]. The above works have proven the effectiveness of structural upgrading other than simple forward strategy, among which the potential of iterative framework is seldom explored. Inspired by the iterative classification method and the iterative AD method proposed by Ma and Chang [25] and Chang et al. [26], we propose a novel iterative framework for HTD, where the iterative DL-based structure is introduced in HTD for the first time. With posterior probability information fed back to the current data matrix for the next iteration, the data quality gets enhanced iteratively. The key problem in designing the iterative framework is the determination of three key elements, the iteration core, the feedback approach, and the termination condition, respectively.

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To reduce the computational complexity and highlight the effectiveness of the latent dimension constraint, the simple structured TAE is chosen as the iteration core. For the feedback approach, it refers to the way how the result obtained in a single iteration is fed back for data quality enhancement. The existing feedback approaches in HSI processing include concatenation [18], weighting [21], and multiplication [17]. In this article, we propose to expand, weigh, and add the CEM-based result to current data pixelwise iteratively for better TD and BS. As the number of iterations increases, the target in the image will be brighter and BKG will be suppressed along with other unwanted signals. And the TI value [27] is designed for adaptive termination judgment in each iteration. The novel method proposed in this article is named the ITAE for HTD. This is the first time an iterative framework with a classic-DL collaborative method for HTD has been introduced, which achieves promising results with a simple structure.

To verify the effectiveness of ITAE, sufficient experiments were conducted in comparison to several state-of-the-art HTD methods under the evaluation of 3-D ROC metrics [28]. Furthermore, the performance of ITAE under noise was evaluated as well as the ablation study. The primary contributions can be summarized as follows.

1) We develop a structurally streamlined and efficient AE-based network for HTD, named TAE. It excels in eliminating noise and interference adaptively by constraining the hidden neurons with estimated VD and driving the network to focus on pertinent target features with KL-based weighting matrix.
2) We introduce a modified KL divergence, named TKL, that enhances target pixel prominence in the reconstruction process, thereby optimizing network parameter updates and improving detection results. The truncated strategy can assure the target-BKG divergence under the potential risk of outliers.
3) Based on TAE, we propose a ITAE for HTD, featuring a feedback loop informed by posterior knowledge and adaptively terminated by TI. This framework includes a weighted rough detection matrix to improve target saliency and enhance target-BKG diversity, further exploiting the potential of TAE's ability for adaptive noise and interference annihilation.
4) The proposed ITAE denotes a novel and general framework for HTD, collaborating traditional HTD techniques and DL theories under an iterative mechanism. With an unsupervised network in an iterative manner, ITAE enables progressive enhancements in TD and BS until optimal results are achieved.

II. Related Works

In this section, recently developed HTD methods in the literature are presented and categorized. Furthermore, to facilitate a better understanding of the proposed methodology, we introduce the classic HTD method CEM [29] theoretically and briefly describe the SDIN model [30].

A. HTD Methods

In past predecessor achievement foundation, extensive methods have been proposed in the field of HTD, which can be broadly classified into three categories: statistical-based methods, decomposition- and representation-based methods, and the emerging ML- and DL-based methods. Specific works are listed and discussed as follows.

1) Statistical-Based Methods: These methods generally rely on the statistical properties of HSIs to detect targets. The basic idea is to measure the distance or distribution similarity between the target signature and the BKG signature statistically. Methods of similarity measurement can be dating back to the typical spectral angle mapper (SAM) [31], Euclidean distance (ED) [32], and spectral information divergence (SID) [33], which are highly interpretable, but lack accuracy. For improvement, some detectors based on statistics and information theory have been proposed, like matched filter (MF) [34], orthogonal subspace minimization (OSP) [35], and CEM [29], along with some improved versions. Their advanced variants have been developed from the perspective of data projection, structural improvement, and feature constraints. For example, the kernel-based CEM (K-CEM) [36] and deep CEM (D-CEM) [8] improve the components' separability by introducing nonlinearity to the detector. The hierarchical CEM (hCEM) [21] and ensemble-based CEM (E-CEM) [18] impose the classic CEM on the hierarchical or ensemble structure, thereby emphasizing the diversity by posterior knowledge. In addition, the sparse-spatial CEM (S-S-CEM) [37] introduces the sparse constraint and spatial regularization to the CEM formulation, generating better detection performance by emphasizing certain physical characteristics. The benchmark methods are of strong generalization ability, while their discriminative ability needs to be enhanced. Their variants improve the methodology of the benchmark algorithms from various perspectives, but their subjection to certain constrains generally introduces new generalization or algorithm complexity problems.

2) Decomposition- and Representation-Based Methods: These methods usually map the original HSIs to a new space obtained by decomposition or representation strategies, where the targets are more discriminative to differentiation.

The decomposition-based methods mainly rely on accurate matrix or tensor decomposition based on the physical characteristics of different components in HSIs and then conduct TD in the decomposed subspace. Related works include the sparse and low-rank matrix decomposition for automatic HTD in [38], the BKG-annihilated OSP in [39], the decomposition model with BKG dictionary learning (DM-BDL) [40], the dictionary learning HTD algorithm based on Tucker tensor decomposition [41], and the intrinsic decomposition-based tensor modeling scheme for HTD [42].

The representation-based methods represent HSIs as a linear combination of a set of basis functions or dictionary atoms based on the physical characteristics of HSIs, including sparse representation, low-rank representation, and collaboration representation. Prior research includes the sparse representation for HTD in [43], the adaptive dictionary construction for TD

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(ADCTD) in [44], regularized tensor representative coefficient model for HTD in [45], the dual-sparsity-constrained (DSC) approach for HTD in [46], and the iterative BKG reconstruction and suppression framework (IBRS) for HTD [24].

These methods are efficient in some scenarios, while they may be sensitive to noise and interference, for the strong reliance they have upon the accurate and adequate generation of representative atoms.

3) ML- and DL-Based Methods: ML- and DL-based methods are drawing increasing attention of scholars because of their capability in deep and nonlinear feature extraction. Free of various assumptions and premises, they can be more generic with untapped potential for better accuracy, which can be typed into methods based on supervised learning, self-supervised learning, and unsupervised learning.

Supervised learning-based methods like [47] use support vector machines (SVMs) [48] to learn to categorize pixels with labeled data. Since it is scarce for HSIs, the corresponding research in HTD is limited considering the range of application.

Self-supervised learning-based methods generate posterior supervised signals by processing unlabeled data for the downstream learning tasks. One of the most representative network is the generative adversarial network (GAN) [49], generating high-fidelity data by playing two neural networks against each other. In 2021, Xie et al. [50] proposed a GAN-based spectral–spatial network, generating the optimal spectral band subset by mapping the relationship between the latent spectral feature space and the original band space. Later, a proposal-weighted two-stage GAN with attention for HTD [51] combines GAN and AE for the extraction of deep discriminative features, of which the sample selection stage addresses the issue of lacking well-labeled training instances. The self-supervised learning-based methods certainly provide a solution to the data-scarce problem in HTD, but the complex network structure and poor interpretability lead to few works in this area.

With no need of labeled data, unsupervised learning algorithms like AE are widely used in HTD. As described in Section I, AE-based methods like DCSSAE [15], HDAE [12], and 3-D-MMRAE [14] improve AE by optimizing network structures and revising loss functions for better detection performance. In addition, the HDAE and subspace projection raised by Shi et al. [23] also use AE for denoising and then applied the denoised image for unsupervised clustering. And the self-supervised learning with deep clustering for TD (SSDCTD) in [52] optimizes the detector with unsupervised network to address the issue of spectral variability. Considerable works have proved the untapped potential that unsupervised learning possesses in HTD. However, as discussed in Section I, there are still some issues to be addressed for detection performance improvement in terms of method efficiency and robustness.

B. Classic CEM

As a classic HTD method, CEM was first proposed in the dissertation of Harsanyi [5] and then further denoted in [29]. It is formulated as a filter design problem, of which the response of desired spectral signature stays put while minimizing the filter output energy. The mean energy of the filter output $\overline{\mathrm{E}}(\delta(\mathbf{X}))$ can be described as

$$\begin{aligned}\overline{\mathrm{E}}(\delta(\mathbf{X})) &= \frac{1}{N}\sum_{i=1}^{N} y_i^2 = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{w}^{\mathrm{T}}\mathbf{r}_i\right)^2 \\ &= \mathbf{w}^{\mathrm{T}}\mathbf{R}\mathbf{w} \end{aligned} \tag{1}$$

where the filter input $\mathbf{X} \in \mathbb{R}^{L\times N}$ refers to the spectral matrix, where L represents the number of spectral bands, N represents the total number of image pixels, and $\mathbf{r}_i \in \mathbb{R}^{L\times 1}$ represent the ith spectral vector in $\mathbf{X}$. And $\mathbf{w} \in \mathbb{R}^{L\times 1}$ is the weighting vector of the filter, generating the filter output $\mathbf{y} = [y_1, y_2, \ldots, y_N]$. $\mathbf{R} \in \mathbb{R}^{L\times L}$ is the correlation matrix of $\mathbf{X}$.

To obtain the optimized weight vector $\mathbf{w}^\star$, the optimization problem is denoted as

$$\begin{aligned}\mathbf{w}^\star &= \arg\min_{\mathbf{w}} \mathbf{w}^{\mathrm{T}}\mathbf{R}\mathbf{w} \\ &\text{subject to } \mathbf{w}^{\mathrm{T}}\mathbf{d} = \mathbf{d}^{\mathrm{T}}\mathbf{w} = 1 \end{aligned} \tag{2}$$

where $\mathbf{d} = (d_1, d_2, \ldots, d_L)^{\mathrm{T}}$ indicates the characteristic vector of interested targets to be detected, and it can be the prior spectrum from the spectral library or the posterior spectrum of the mean spectrum from certain pixels in CEM detection.

To solve this optimization problem, we introduce the Lagrange multiplier, which is used to construct the Lagrangian objective function. After derivation, the optimal solution can be calculated as follows:

$$\mathbf{w}^\star = \frac{\mathbf{R}^{-1}\mathbf{d}}{\mathbf{d}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{d}}. \tag{3}$$

And the detection result of $\mathbf{X}$ is given by $\mathbf{w}^\star\mathbf{X}$.

C. SDIN Model

HSIs capture the spectral signatures of all the components within the scene. In the SDIN linear mixing model proposed by Du and Chang [30], an HSI is modeled as a linear superposition of the targets, BKG, interference, and noise, which can be expressed as

$$\mathbf{r} = \mathbf{S}\boldsymbol{\alpha} + \mathbf{L}\boldsymbol{\beta} + \mathbf{I}\boldsymbol{\gamma} + \mathbf{N} \tag{4}$$

where $\mathbf{S} \in \mathbb{R}^{L\times n_t}$, $\mathbf{L} \in \mathbb{R}^{L\times n_b}$, $\mathbf{I} \in \mathbb{R}^{L\times n_i}$, and $\mathbf{N} \in \mathbb{R}^{L\times 1}$ represent the interested target spectral matrix, the BKG spectral matrix, the interference spectral matrix, and the noise vector, respectively. And $\boldsymbol{\alpha} \in \mathbb{R}^{n_t\times 1}$, $\boldsymbol{\beta} \in \mathbb{R}^{n_b\times 1}$, and $\boldsymbol{\gamma} \in \mathbb{R}^{n_i\times 1}$ represent the abundance of target, BKG, and interference, respectively.

In HTD, the target refers to the specific type of objects to be identified, and the BKG refers to other undesired targets in the image. Interference and noise are two types of unwanted signals that may significantly affect the detection results. Noise is generally random and unpredictable, while interference is more specific and deterministic. In this article, we propose to exploit the statistical characteristics of different components in HSIs to enhance the conventional unsupervised DL method, AE, for improved detection performance.

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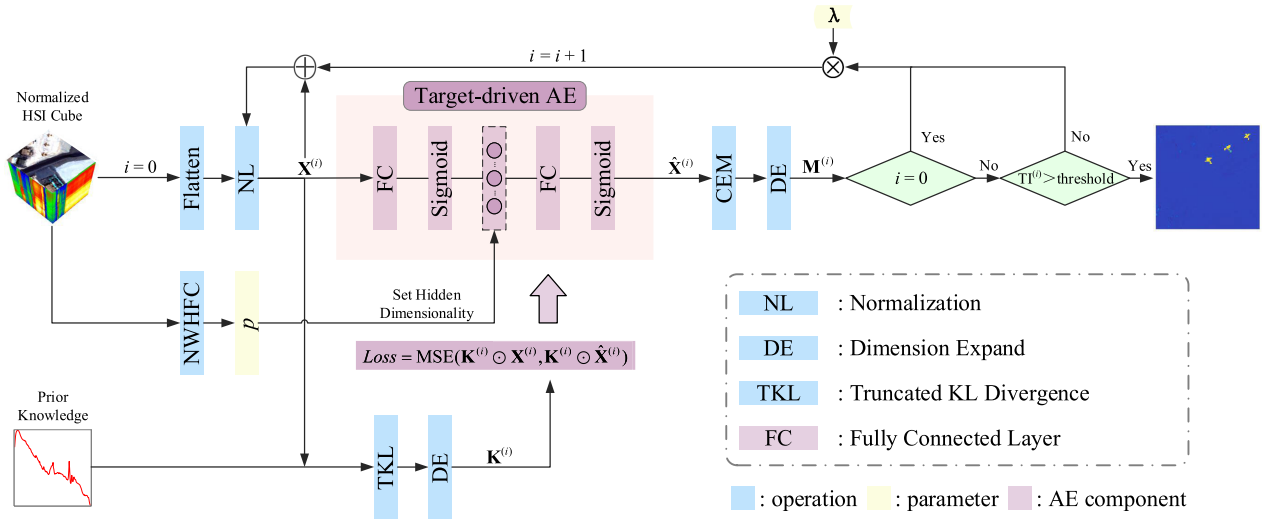


Fig. 1. Overall flowchart of ITAE for HTD.

III. Methodology

This section presents a comprehensive analysis of ITAE for HTD. Fig. 1 shows the overall flowchart of the proposed ITAE framework. First, the hyperspectral cube is transformed into a 2-D matrix without damaging the spatial and spectral integrity of the data. Then, the flattened HSI matrix is normalized and sent to the TAE, where the dimension of latent code is constrained by the estimated VD of the original HSI cube to suppress interference and noise. In addition, to drive the reconstruction prior to the target pixels over the BKG pixels, the mean square error (MSE) loss is multiplied by a weighting matrix calculated by the proposed TKL. Then, the reconstructed data are applied to CEM for coarse target detection and dimensional expansion, where the map is replicated and scaled from a 1-D vector to a 2-D matrix, to a weighting matrix, which is utilized for data refinement in the next iteration. The iteration terminates when the TI value surpasses the preset threshold, and the final detection map is the last CEM map.

A. Target-Driven AE

In this article, we introduce an enhanced TAE for HTD, which employs an adaptive dimensionality constraint in the hidden layer, which encourages the network to focus on essential features of the targets and BKG, while disregarding extraneous information. Furthermore, the novelty designed loss function for the TAE is a weighted sum of pixelwise reconstruction errors. The weights are derived from the proposed TKL, which prioritizes the reconstruction accuracy of pixels with spectral profiles similar to the targets. This method improves reconstruction efficiency by deemphasizing the learning of BKG details. Consequently, the resultant reduction in noise and interference, along with a smoother reconstructed BKG, significantly aids in distinguishing the BKG from the targets in subsequent detection tasks.

1) Adaptive Dimensionality Constraint: Typically, an AE network encodes the input into a lower dimensional hidden space and then decodes the latent feature back to the original input space. The number of latent neurons decides how much information is compressed into the encodings. A larger size yields a more faithful reconstruction, while a smaller size leads to reconstruction that highlights significant patterns. By constraining the hidden dimensionality, the AE can be driven to prioritize the prominent features of the input data while compressing and reconstructing. It has to discard details of less importance to meet the size restriction.

In HTD, according to the SDIN decomposition model described in Section II-C, an HSI can be considered a combination of four components, BKG, targets, noise, and interference. Compared to other two components, the BKG and targets clearly take the decisive role with the largest proportion in an image. And the interference and noise components are two kinds of unwanted signals and usually decrease the detector performance, which need to be eliminated. Instead of representing statistics or features of noise and interference, which is usually unreliable because of their uncertainty, we propose an adaptive method for noise and interference elimination in the process of AE training. To utilize the latent dimensionality of AE to influence the reconstruction priorities, we estimated the sum of the dimensions of the targets and BKG components, using it to constrain the hidden dimensionality. By setting the hidden dimensionality of TAE as the estimated sum dimensionality of the targets and BKG, the network is driven to take advantage of limited information capacity to learn the discriminative features of the targets and BKG during encoding. Therefore, by decoding the latent features back into the original data space, the reconstructed data can maintain the details of the targets and BKG, while neglecting the noise and interference, generating an image of higher quality for the downstreaming detection.

To determine the sum dimensionality of the target and BKG components in HSIs adaptively, we have investigated the intrinsic physical characteristics of the four components described in (4). The targets and the BKG, known as, the uninterested targets, are the fusion of distinct spectral signal sources in HSIs, which are denoted as endmembers.

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Differently, the formation of noise and interference can not be represented by the combination of endmembers multiplied abundance, for their appearances are often caused by mechanical error, manual error, or meteorological reasons. Therefore, the sum dimensionality of the target and BKG components can be estimated with the number of spectrally distinct signal sources in HSIs, which is also the number of endmembers. Based on this assumption, we scale endmember number estimation methods to the application of estimating the dimensionality of discriminative features in HSIs to constrain the unsupervised network for prioritized reconstruction.

In this article, we choose the NWHFC method for the estimation of number of distinct signal sources in HSIs. As an endmember number estimation algorithm, NWHFC is computationally efficient and robust to noise. As denoted in [16], NWHFC is developed based on the HFC method with a noise whitening fore-step to improve its robustness to noise. The derivation process can be found in [16] and [53]. Here, we use it for noise and interference suppression in AE, implementing two steps: noise whitening and VD estimation. The purpose of noise whitening is to eliminate the influence of noise on the eigenvalues of data's correlation matrix and covariance matrix. To achieve this, we have to estimate the covariance matrix of the original data. For an input spectral matrix $\mathbf{X}$, the correlation matrix $\mathbf{R}$ and covariance matrix $\mathbf{K}$ are calculated

$$\begin{aligned}\mathbf{R} &= \frac{1}{N}\mathbf{X}\mathbf{X}^{\mathrm{T}} \\ \mathbf{K} &= \frac{1}{N-1}\left(\mathbf{X}-\bar{\mathbf{X}}\right)\left(\mathbf{X}-\bar{\mathbf{X}}\right)^{\mathrm{T}}\end{aligned} \tag{5}$$

where $\bar{\mathbf{X}}$ refers to the average spectrum of the input HSI data, which is calculated as $\frac{1}{N}\sum_{i=1}^{N}\mathbf{X}(:,i)$, and $\mathbf{X}(:,i)$ denotes the ith column of the input spectral matrix. Since the assumed noise is uncorrelated, its covariance matrix is ought to be a diagonal matrix. As described in [16], the estimated noise covariance matrix $\mathbf{K}_{\mathrm{n}}$ is the inverse of the diagonal matrix of the inverse of the data's covariance matrix

$$\mathbf{K}_{\mathrm{n}} = \left(\mathrm{diag}\left(\mathbf{K}^{-1}\right)\right)^{-1} \tag{6}$$

where the function diag$(\cdot)$ is an operation that takes a matrix as input and returns a matrix of the same size, where the diagonal elements of the output matrix are the same as the diagonal elements of the input matrix, and all other elements are zero. The estimated noise covariance matrix is then used to whiten the data, which is denoted as

$$\mathbf{X}_{\mathrm{w}} = \mathbf{K}_{\mathrm{n}}^{-\frac{1}{2}}\mathbf{X}. \tag{7}$$

The noise-whitened data $\mathbf{X}_{\mathrm{w}}$ are then used for VD estimation. First, the correlation matrix $\mathbf{R}_{\mathrm{w}}$ and covariance matrix $\mathbf{K}_{\mathrm{w}}$ of the noise-whitened data are calculated, which generates two sets of eigenvalues $\{\gamma_1 \geqslant \gamma_2 \geqslant \cdots \geqslant \gamma_L\}$ and $\{\nu_1 \geqslant \nu_2 \geqslant \cdots \geqslant \nu_L\}$, respectively. The eigenvalues are a measure of variance or the amount of energy gathered in each direction. Based on the assumption that signal is nonrandom and positive constant, if there is a positive value of the signal contributing in a particular direction rather than noise, we can have $\gamma_l > \nu_l$. Hence, we calculate the difference between the eigenvalues and generated the difference matrix

$$\mathbf{D} = \begin{bmatrix} \gamma_1-\nu_1 & 0 & \cdots & 0 \\ 0 & \gamma_2-\nu_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \gamma_L-\nu_L \end{bmatrix}. \tag{8}$$

Ideally, the estimated VD value is the rank of $\mathbf{D}$, under the assumption that the total number of samples N is sufficiently large. However, in real physical condition, the range of N is limited, so we need a predefined false alarm rate to improve the accuracy of VD estimation. The false alarm rate is the probability of a false positive, which is an event where the algorithm predicts a VD when there is none. The threshold value τ_l for band l is calculated based on the preset false alarm rate ρ using the theory of multiple regression analysis

$$\tau_l = \sqrt{2} \times \frac{2\sqrt{\gamma_l^2+\nu_l^2}}{N} \times \mathrm{erfinv}(1-2\times\rho) \tag{9}$$

where function erfinv$(\cdot)$ refers to the inverse error function, which is utilized to calculate the cumulative distribution function of the normal distribution.

Then, the estimated VD value p is given by

$$p = \sum_{i=1}^{L}\sum_{j=1}^{L}\left[D_{i,j} > 0\right] \tag{10}$$

where $[D_{i,j} > 0]$ means that when $D_{i,j} > 0$, the value is 1; otherwise, the value is 0. The estimated VD value p is used for the constraint of AE, where the dimensionality of latent code is set to this specific value. By adaptively annihilating noise and interference that may damage the subsequent detection, the network is forced to perceive the components that need to be preferably attended to.

2) Target-Biased Reconstruction: The traditional loss function of AE is the MSE between the input and output, measuring the reconstruction accuracy by similarity evaluation. In HTD, compared to the BKG, noise, and interference, the characteristics of the target are relatively more crucial for the subsequent detection. With the same training epochs, the details of the target component, especially the edge details, need to be prioritized. Especially, in some small target scenarios, it usually occurs that information of the target is neglected for better retaining the BKG details, which is clearly unsatisfactory. Therefore, the AE for HTD is expected to be prone to the reconstruction of the targets for the downstreaming detection task. To achieve this, a TKL-weighted loss function for TAE is proposed, and the flowchart of obtaining TKL is shown in Fig. 2.

KL divergence is a measurement for differentiating two probability distributions. It has applications in various fields, such as information theory, ML, and statistics, and is commonly used in tasks such as model selection, feature selection, and clustering. As a probability similarity evaluation criterion, intuitively, KL divergence measures the spectral shape similarity, ignoring the amplitude stretch. This provides us with the robust posterior knowledge that is inclusive of the spectral

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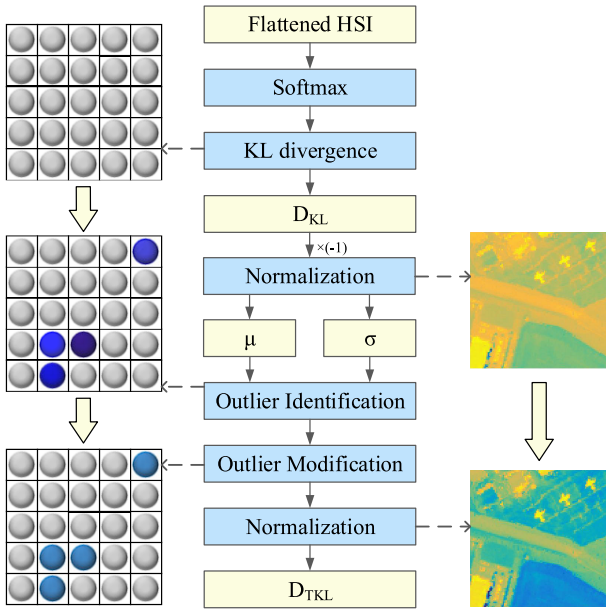


Fig. 2. Calculation of TKL.

variability. However, in HTD, when calculating the KL diversity between the pixels in HSIs and the target spectrum, it can be found that the overall distribution of KL map is sensitive to several extreme outliers. As shown in Fig. 2, the diversity of targets and BKG is decreased because of several dark blue points after normalization. To improve the distribution of KL divergence and the diversity of the targets and BKG, we propose an improved version of KL divergence in HTD, the TKL, which is used by the TAE loss function. The calculation of TKL is as follows.

First, instead of using probability distributions as input, spectral vector of pixels in HSIs and prior spectrum is sent for KL divergence calculation after preprocessing

$$\mathbf{p}_{i,j} = \frac{e^{\mathbf{X}_{i,j}}}{\sum_{k=1}^{L} e^{\mathbf{X}_{k,j}}}$$
$$\mathbf{q}_i = \frac{e^{\mathbf{d}_i}}{\sum_{k=1}^{L} e^{\mathbf{d}_k}}$$
$$i = 1, 2, \ldots, L \text{ and } j = 1, 2, \ldots, N. \tag{11}$$

This preprocessing procedure is to ensure each spectrum conforms to the characteristics of a probability distribution, which is summing to 1. And the KL divergence of the jth column of the spectral matrix $\mathbf{X}$, $D_{\mathrm{KL}}(j)$, is calculated as

$$D_{\mathrm{KL}}(j) = \sum_{i=1}^{L} \mathbf{q}_i \log \frac{\mathbf{q}_i}{\mathbf{p}_{i,j}} \tag{12}$$

where $D_{\mathrm{KL}}(j)$ refers to the jth element in the distribution distance vector $\mathbf{D}_{\mathrm{KL}} \in \mathbb{R}^{1\times N}$. The smaller the value, the more similar the pixel is to the target spectrum, indicating a higher likelihood of being the target. To make the vector positively correlated with the physical similarity, we take its negative and then normalize it. Since extreme values can have a significant impact on the overall distribution of $\mathbf{D}_{\mathrm{KL}}$, causing most of the values to be concentrated in a small area, the discrimination between BKG and target may be reduced. To address this issue, we utilize the truncated strategy to identify the extreme outliers and then modify them. Motivated by the 3-σ limits statistical calculation criterion, which refers to the qualified data lay within three standard deviations from a mean, we also evaluate data quality using three standard deviation and the mean values. Aiming to improving the diversity of the BKG and targets, outliers are identified as data points that fell below the mean minus three standard deviations and are subsequently truncated to marginal values

$$\mu = \frac{1}{N}\sum_{j} D_{\mathrm{KL}}(j)$$
$$\sigma = \sqrt{\frac{1}{N-1}\sum\nolimits_{j} (D_{\mathrm{KL}}(j) - \mu)^2}. \tag{13}$$

Find the values that are bigger than $\mu + 3\sigma$ and set them to the specific value of $\mu + 3\sigma$

$$s_j = \begin{cases} \mu + 3\sigma, & \text{if } D_{\mathrm{KL}}(j) > \mu + 3\sigma \\ D_{\mathrm{KL}}(j), & \text{otherwise.} \end{cases} \tag{14}$$

Then, the revised vector is normalized

$$s_j = \frac{s_j}{\sum_k s_k} \tag{15}$$

where s_i refers to the ith element in the truncated vector $\mathbf{S}$ of length N. To guide the TAE to prioritize the reconstruction of pixels that are more similar to the target, the loss function is set to the Hadamard product of the traditional MSE loss and the TKL $\mathbf{T}$, which is calculated as

$$\mathbf{T} = \mathbf{US} \tag{16}$$

where $\mathbf{U} = [1, 1, \ldots, 1] \in \mathbb{R}^{L\times 1}$, and $\mathbf{T}$ is an $L \times N$ weighting matrix, generating the MSE loss between the weighted input of TAE $\mathbf{X}$ and the reconstructed data $\hat{\mathbf{X}}$

$$\mathrm{Loss} = \mathrm{MSE}\left(\mathbf{T} \odot \mathbf{X}, \mathbf{T} \odot \hat{\mathbf{X}}\right) \tag{17}$$

where the symbol $\odot$ refers to the Hadamard product operation, which computes the elementwise product of the corresponding elements in the two input vectors to produce the output matrix. This way, the TAE prioritizes the reconstruction of the target component rather than the BKG component discussed in Section II-C.

B. ITAE Framework

For the TAE introduced in Section III-A, although the subsequent detection results can be improved by improving the data quality, the effect is limited due to the inadequate use of posterior knowledge. To further improve the detection effectiveness, we introduce the iterative framework to TAE, which is identified as ITAE. And Algorithm 1 provides its implementation details.

In Section III-A, the output of the TAE can be obtained as $\hat{\mathbf{X}} \in \mathbb{R}^{L\times N}$. For the downstreaming detection task, the CEM detector is utilized for rough target detection. The input of CEM is the reconstructed data $\hat{\mathbf{X}}$ and the prior knowledge $\mathbf{d}$, according to [29], and the target map can be calculated as

$$\mathbf{F} = \frac{\mathbf{d}^{\mathrm{T}}\hat{\mathbf{R}}^{-1}\hat{\mathbf{X}}}{\mathbf{d}^{\mathrm{T}}\hat{\mathbf{R}}^{-1}\mathbf{d}} \tag{18}$$

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Algorithm 1 ITAE Framework

Input: Flattened and normalized HSI matrix $\mathbf{X} \in \mathbb{R}^{L\times N}$, the normalized prior spectrum $\mathbf{d}$, the tradeoff parameter for updating AE input λ, the stop criterion threshold ϵ.

Initialize $\mathbf{X}^{(0)} = \mathbf{X}$, $\mathbf{TI}^{(0)} = 0$, $\mathbf{M}^{(0)} = \mathbf{0}_{L\times N}$, $i = 1$.
Set the dimensionality of latent code in the target-driven AE via (10).
while $\mathbf{TI}^{(i-1)} \leq \epsilon$ and $i \leq 8$ **do**
    Calculate the Truncated KL Divergence via (16).
    Set the loss function of the target-driven AE via (17).
    Train the target-driven AE with input of $\mathbf{X}^{(i-1)}$.
    Obtain the reconstructed data $\hat{\mathbf{X}}$.
    $\mathbf{F} = \frac{\mathbf{d}^T\hat{\mathbf{R}}^{-1}\hat{\mathbf{X}}}{\mathbf{d}^T\hat{\mathbf{R}}^{-1}\mathbf{d}}$, $\hat{\mathbf{R}} = \frac{1}{N}\hat{\mathbf{X}}\hat{\mathbf{X}}^T$.
    $\mathbf{M}^{(i)} = \mathbf{UF}$, $\mathbf{U} = [1, 1, \ldots, 1] \in \mathbb{R}^{L\times 1}$.
    $\mathbf{X}^{(i)} = \mathbf{X}^{(i-1)} + \lambda\mathbf{M}^{(i)}$.
    $\mathrm{TI}^{(i)} = \frac{|\mathbf{M}^{(i)}\cap\mathbf{M}^{(i-1)}|}{|\mathbf{M}^{(i)}\cup\mathbf{M}^{(i-1)}|}$.
    $i = i + 1$.
end while
$\mathbf{TM} = \mathbf{F}$.

Output: Targetmap in the last iteration, **TM**.

where $\hat{\mathbf{R}}$ refers to the correlation matrix of $\hat{\mathbf{X}}$ and $\mathbf{F} \in \mathbb{R}^{1\times N}$. To align the dimensions of the target map and the AE's input for the iterative data quality enhancement, dimension expansion is performed

$$\mathbf{M} = \mathbf{UF}. \tag{19}$$

Then, the first iteration of target detection result has been obtained with the input of the original flattened HSI data. For the next iteration, the input of the network is updated with the weighted sum of $\mathbf{X}$ and the expanded target map $\mathbf{M}$ with the tradeoff parameter λ

$$\mathbf{X} = \mathbf{X} + \lambda\mathbf{M}. \tag{20}$$

Here, with updated input, the reconstruction and detection process are conducted according to (5)–(20).

The stop criterion of the framework is determined by the value of TI, also known as the Jaccard index [27], which is a measure of similarity between two adjacent maps. We use the TI value of CEM target maps in two adjacent iterations to determine whether the stop condition has been reached. If the value reaches a certain threshold, it is considered that the iteration has little effect on the detection and the result is reliable, and then, the iteration can be stopped. The TI value in the nth iteration is calculated as

$$\mathrm{TI}^{(n)} = \frac{|\mathbf{M}^{(n)} \cap \mathbf{M}^{(n-1)}|}{|\mathbf{M}^{(n)} \cup \mathbf{M}^{(n-1)}|} \tag{21}$$

where the superscript n represents the number of iterations. The threshold for the stop criterion is preset based on the number of pixels in the image.

IV. Experimental Analysis

In this section, we evaluate the effectiveness of the proposed ITAE framework using six real hyperspectral datasets with varying scenarios. The evaluation will be conducted using 3-D ROC curves, and parameter analysis will be performed to discuss the influence of the tradeoff parameter. The algorithm's performance will be compared with seven other state-of-the-art algorithms, focusing on detection accuracy, robustness with noise, and ablation analysis.

A. Dataset Description and Comparative Methods

Datasets used in this article include the Urban dataset, the Abu-urban-1 dataset, the Abu-urban-2 dataset, the Sandiego dataset, the Abu-airport dataset, and the Abu-beach dataset, covering three different scenes of urban, airport, and beach. In addition, the datasets were collected by three different sensors, including the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor, the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, and the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor. Furthermore, targets in the HSIs conclude vehicles, buildings, airplanes, and other man-made objects with different sizes. In these datasets, the noisy bands and water-absorption bands are removed, respectively. More concretely, the detailed information of each dataset are established in Table I, where the image size refers to the number of pixels obtained by multiplying the width and height of the HSI image on a pixel-by-pixel basis, and the size of the target refers to the number of target pixels. Fig. 3 shows the visualization of six HSI datasets, where the pseudocolor image of each hyperspectral cube is formed by the first band, the median band, and the last band. Notably, the third row in Fig. 3 shows the average spectra of the target and BKG components.

To verify the effectiveness of the proposed ITAE method, we selected seven representative and state-of-the-art methods for comparison. Considering the variations in HTD methods, we chose benchmarks corresponding to the four types of HTD discussed in Section II-A, including CEM [29], hCEM [21], E-CEM [18], decomposition-based dictionary learning (DM-BDL) [40], BKG annihilated orthogonal subspace projection (BA-OSP) [39], DSC method [46], and BKG learning based on target suppression constraint for HTD (BLTSC) [54]. Among these seven benchmarks, we included DM-BDL [40], which is based on dictionary construction, to support our innovations of substituting the dictionary construction with probability-based measurements. In addition, the BLTSC [54] is also based on AE reconstruction, serving as a representative of the AE-based HTD methods. Furthermore, to validate the effectiveness of utilizing the posterior knowledge to enhance the detection performance with iteration strategy, hCEM [21] and E-CEM [18] are chosen as representatives with hierarchical or ensemble techniques.

Notably, respectively, authors kindly shared the implementation code of comparison methods utilized in this article, and the specific parameter settings for these methods are listed in Table II. We conduct all experiments with Python 3.7 and MATLAB R2022b on Windows 10 with CPU Intel Core i9-10980HK at 2.40 GHz and GPU NVIDIA GeForce RTX 2080 Super, 8-GB VRAM.

B. Evaluation Metrics

A comprehensive and objective evaluation metrics is crucial for algorithm design, optimization, and comparison. The

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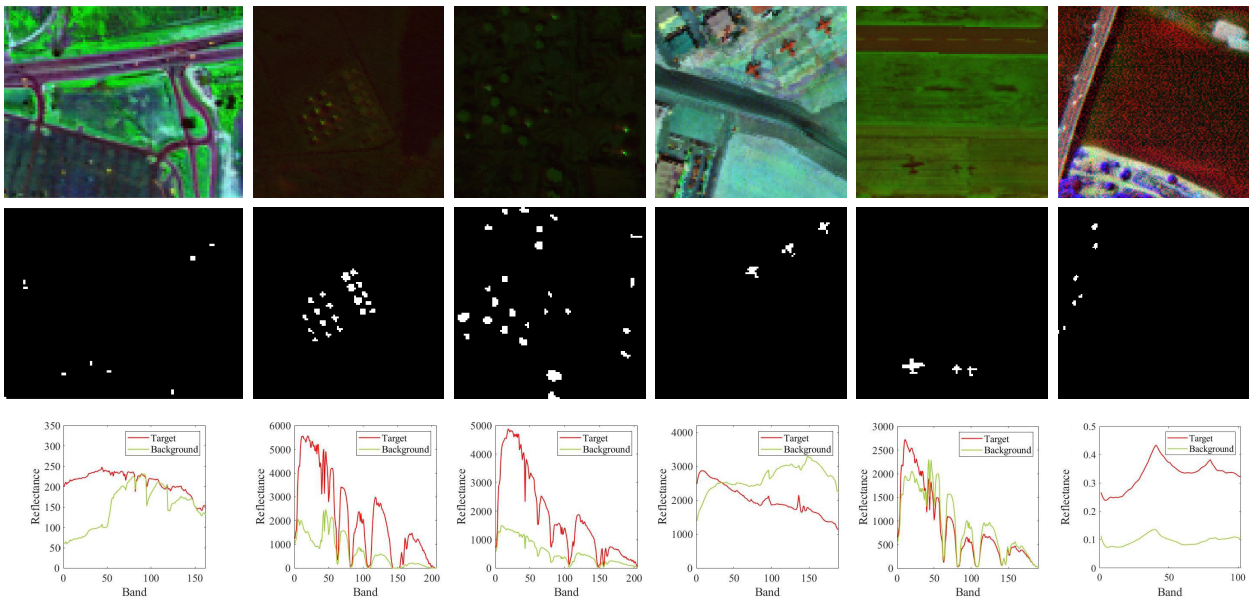


Fig. 3. Visualization of six HSI datasets. The first row from top to bottom shows the pseudocolor images of the datasets, the second row shows the ground-truth maps, and the third row shows the average spectra of BKG and targets. From left to right, they refer to the Urban dataset, the Abu-urban-1 dataset, the Abu-urban-2 dataset, the Sandiego dataset, the Abu-airport dataset, and the Abu-beach dataset, respectively.

TABLE I
DETAILED INFORMATION OF SIX HSI DATASETS

Dataset	Sensor	Location	Spatial Resolution (m)	Image Size	Spectral Channels	Target	
						Type	Size
Urban	HYDICE	Michigan	2.0	80*100	162	Vehicle	21
Abu-urban-1	AVIRIS	Texas	17.2	100*100	207	Building	155
Abu-urban-2	AVIRIS	Los Angeles	7.1	100*100	205	Building	272
Sandiego	AVIRIS	San Diego	3.5	100*100	189	Plane	57
Abu-airport	AVIRIS	Gulfport	3.4	100*100	191	Plane	60
Abu-beach	ROSIS-03	Pavia	1.3	150*150	102	Vehicle	68

TABLE II
DETAILED PARAMETER SETTINGS FOR COMPETITIVE METHODS

Methods	Parameter Settings
hCEM [21]	Adjustment parameter: 200; Tolerance: 1e-3
E-CEM [18]	Windowsize: $[1/4, 2/4, 3/4, 1] \times Bandsize$ Number of layers: 5 Number of CEM detectors: 6 Regulation parameter: 10^{-6}
DM-BDL [40]	Number of background atoms: 20 Learning parameter: 20 Decay parameter: 0.95 Tradeoff parameter $\lambda : 0.01, \beta : 0.1$
BA-OSP [39]	False alarm rate: 10^{-3}
BLTSC [54]	Training data ratio: 0.75 Dimensionality of z: 50 Number of epoches: 500 Learning rate: 10^{-4} Binarization threshold: 0.15
DSC [46]	Number of target/background atoms: 25/10
ITAE	Learning rate: 10^{-3}; Batch size: 64 Number of epochs: 500 Max number of iterations: 8 Stop criterion threshold: 0.999

commonly used evaluation metrics for HTD contains indicators like precision, recall, and F1-score, measuring the detection ability and identification ability of the detector. For more concrete evaluation, the 2-D ROC curve with the false alarm rate P_F on the horizontal axis and the detection rate P_D on the vertical axis can also reflect the robustness and stability of the detector by observing the curve shape and calculating the area under the curve, which is AUC value.

Though the 2-D ROC curve and derived AUC value show great performance on detection measure, there still exists some limitations. First, the ROC curve is derived from an NPD to find (P_D,P_F) on a varying threshold τ, which should be a discrete value. However, in the 2-D ROC curve generation, the interpolation method is used for curve smoothing, resulting in the obtained curve and AUC value inaccurate. In addition, the P_D and P_F determined by the same threshold τ are not independent and one affects the other. Hence, the 2-D ROC curve can only be used to evaluate the detector effectiveness, but not TD or BS. Lastly, since two different sets of (P_D,P_F) may generate the same AUC value, the traditional AUC can also be limited in detector performance evaluation.

To tackle the above limitations of 2-D ROC curve and achieve more comprehensive evaluation, the improved 3-D ROC curve and derived indicators described in [28] are utilized for more effective evaluation for detection algorithms. By taking the other variable, threshold τ, into consideration,

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TABLE III
3-D ROC CURVE-DERIVED EVALUATION METRICS

Category	Indicator	Definition	Indicator Meaning
Target Detectability, TD	$AUC_{(D,\tau)}$	$AUC_{(D,\tau)} \in [0,1]$	The AUC value of a (P_D, τ) curve, reflecting the target detectability.
	AUC_{TD}	$AUC_{TD} = AUC_{(D,F)} + AUC_{(D,\tau)} \in [0,2]$	The joint target detectability to calculate the effectiveness and TD of a detector.
Background Suppressibility, BS	$AUC_{(F,\tau)}$	$AUC_{(F,\tau)} \in [0,1]$	The AUC value of a (P_F, τ) curve, reflecting the background suppressibility.
	AUC_{BS}	$AUC_{BS} = AUC_{(D,F)} + AUC_{(F,\tau)} \in [-1,1]$	The joint background suppression (BS) of a detector.
Overall Performance	$AUC_{(D,F)}$	$AUC_{(D,F)} \in [0,1]$	The AUC value of a (P_D, P_F) curve, reflecting the detector effectiveness.
	AUC_{TDBS}	$AUC_{TDBS} = AUC_{(D,\tau)} - AUC_{(F,\tau)} \in [-1,1]$	The joint target detectability and background suppression (TDBS) of a detector.
	AUC_{SNPR}	$AUC_{SNPR} = \frac{AUC_{(D,\tau)}}{AUC_{(F,\tau)}} \in [0,+\infty)$	The signal-to-noise probability ratio (SNPR) of a detector.
	AUC_{ODP}	$AUC_{ODP} = AUC_{(D,F)} + AUC_{(D,\tau)} - AUC_{(F,\tau)} \in [-1,2]$	The overall detection probability (ODP) of a detector.

the generated 3-D curve with eight derived AUC indicators can analyze the TD, BS, and effectiveness of the detection algorithm independently. Table III shows a detailed description of the eight AUC values.

C. Sensitivity Analysis

In the ITAE framework, the hyperparameters preset in the detection process can be categorized into two groups: those for training and those for iteration processes. Training hyperparameters include the training batch size, learning rate, and number of epochs, which are determined using a grid search method with candidate values [64, 128, 256], $[10^{-4}, 5\times 10^{-4}, 10^{-3}, 5\times 10^{-3}, 10^{-2}]$, and [50, 1000] on the Urban dataset, and the specific range is empirical. The determined optimal hyperparameters, which include a batch size of 64, a learning rate of 10^{-3}, and 500 training epochs, are consistently applied across all training procedures on the other five datasets to maintain uniformity. In particular, the number of hidden neurons is a not predefined parameter but calculated by the NWHFC method [16], varying among different scenarios.

Predefined iterative hyperparameters include the tradeoff parameter λ, the max number of iterations, and the termination criterion threshold. Among them, the optimal choices of the max number of iterations and the termination criterion threshold are primarily based on experimental results across six datasets. In addition, as shown in (20), the weighting parameter λ governs the extent to which the CEM detection result on the HSI is enhanced. This process highlights pixels that are likely to belong to targets and suppresses those that are likely to belong to the BKG. The optimal value of λ depends on the specific application and the tradeoff between detection accuracy and computational efficiency. A smaller value of λ leads to a gentler revision of the image quality, which may require more iterations to achieve a satisfactory detection result, thereby increasing the running time. Conversely, a larger value of λ may excessively alter the image, resulting in more falsely detected pixels.

To analyze the detection performance across six datasets under different λ, we chose the overall detection probability (ODP) indicator for evaluation. We selected an interval of 0.1–2.0 for λ, with an interval of 0.1. The ODP under varying λ is shown in Fig. 4, with the marker on each curve indicating the optimal choice of λ. The optimal λ for dataset Urban, Abu-urban1, Abu-urban2, Sandiego, Abu-airport, and Abu-beach is 1.4, 1.7, 1.5, 1.5, 1.3, and 1.4, respectively.

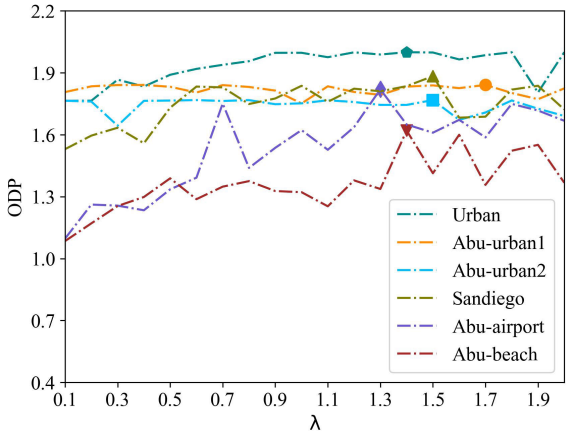


Fig. 4. Impact of parameter λ on ODP across six datasets. The markers on each curve indicate the maximum ODP value for the respective dataset.

Fig. 4 indicates that ODP generally increases with λ up to a certain optimal point, beyond which it either plateaus or slightly decreases, indicating a potential overfitting scenario. This behavior underlines the model's sensitivity to the tradeoff parameter and suggests an optimal range for λ that maximizes detection performance, which is [1.3, 1.7]. Notably, the optimal λ value is dataset-specific, indicating that our model is adaptable to the uniqueness of each dataset, which is a critical consideration in HTD applications. Furthermore, the variations in ODP across different datasets at varying λ values also reflect the model's ability to handle dataset variations and uncertainties effectively. For instance, while the Urban dataset showed stable performance over a wide range of λ values, others like Abu-beach and Abu-urban-2, where the BKG is more complex, exhibited more pronounced fluctuations. The Abu-airport dataset showed a consistently increasing trend

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in ODP with larger λ values, suggesting a lower sensitivity to overfitting or an inherent dataset characteristic that favors a more pronounced emphasis on target features. This demonstrates the model's responsive behavior to different data characteristics. Comprehensively, within the suggested range of λ values, ITAE performs well in all six datasets despite the variations, giving the convincing conclusion that ITAE suits well in varying scenes.

D. Detection Performance Analysis

Based on the parameter settings outlined in Table II and Section IV-C, we conducted a series of competitive experiments using visualization, qualitative, and quantitative analysis. Specifically, for the proposed ITAE method, we adaptively determined the hidden dimensionality of the latent space in AE using (10). For the six HSI datasets, we set the hidden dimensionality to 16, 17, 17, 11, 17, and 7, respectively. Moreover, the number of iterations for each dataset varies from the TI values in (21), which is 4, 5, 7, 8, 5, and 8, respectively.

For enhanced clarity and persuasiveness, we separately illustrate the impact of iteration times on ITAE performance with the ROC-derived AUC results. The crucial measurements of TD, BS, and ODP are chosen for analyzing the impact of iteration times on ITAE performance, and the illustration is shown in Fig. 5. And the final detection results are illustrated in the form of detection maps in Fig. 6 and 3-D ROC-derived AUC values in Tables IV–IX.

The visualization results of detection target maps are shown in Fig. 6. It shows that the targets in the ITAE target map are more prominent, while the BKG is smoothed and suppressed more thoroughly, indicating better performance compared to other comparative methods.

Quantitatively, 3-D ROC-derived AUC results of competitive methods for six datasets are shown in Tables IV–IX, evaluating the detection performance in the perspectives of TD, BS, and overall performance. Notably, evaluation metrics of the last iteration in ITAE are established for more comprehensive evaluation.

1) TD: The values of TD indicators prove that the ITAE outperforms other methods in terms of highlighting targets, as evidenced by the larger $\mathrm{AUC}_{(\mathrm{D},\tau)}$ and $\mathrm{AUC}_{\mathrm{TD}}$ values obtained for all six datasets. More concretely, for datasets Abu-urban-1 and Abu-urban-2, the first iteration of ITAE can achieve better TD results among other methods. And in dataset Urban, the $\mathrm{AUC}_{(\mathrm{D},\tau)}$ and $\mathrm{AUC}_{\mathrm{TD}}$ in the last iteration achieve the maximum threshold of 1.0 and 2.0, emphasizing the outstanding performance of ITAE in TD. Relatively, the hCEM, E-CEM, and BA-OSP also display satisfactory performance in target highlight, followed by BLTSC and CEM, and DM-BDL and DSC may need further optimization in TD.

2) BS: The values of BS indicators show that ITAE is also efficient in annihilating the unwanted components. Under datasets Urban and Abu-urban-2, the $\mathrm{AUC}_{(\mathrm{F},\tau)}$ and $\mathrm{AUC}_{\mathrm{BS}}$ values are both either optimal or suboptimal, especially the $\mathrm{AUC}_{(\mathrm{F},\tau)}$ for dataset Urban is 0.25 times that of the suboptimal, generated by BLTSC. For dataset Abu-urban-1, Sandiego, and Abu-beach, at least one BS metric is optimal or suboptimal, followed by dataset Abu-airport-4, which have more room for improvement in BS. Since there exists a tradeoff relationship between the TD ability and BS ability, considering the targets are strengthened prominently, the BS ability that ITAE shows is as well promising. In comparative methods, BLTSC and DSC establish better performance and BA-OSP shows relatively weak performance in BS.

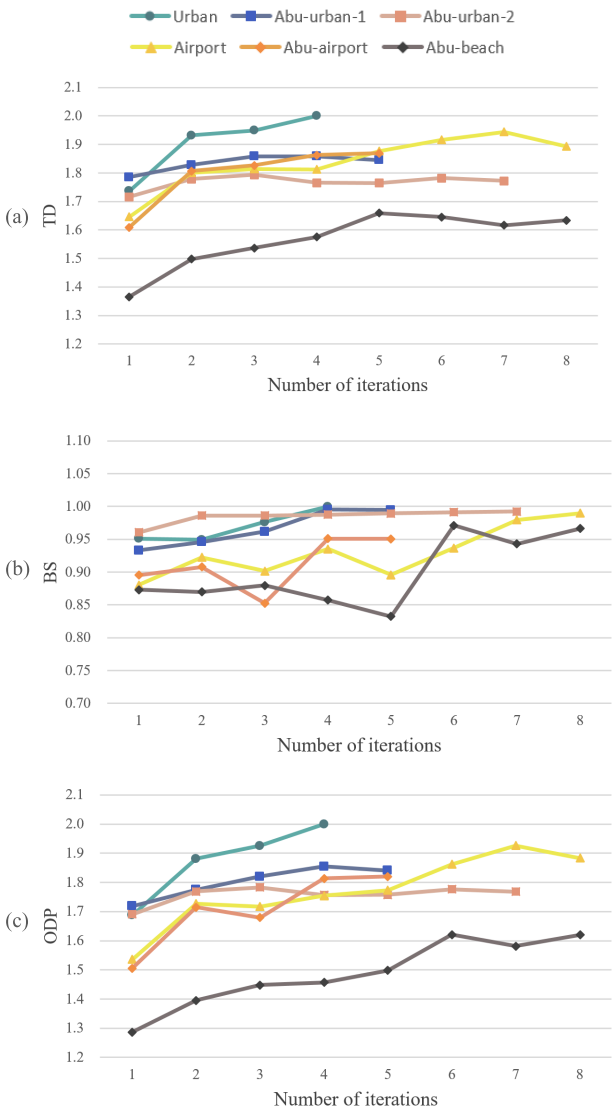


Fig. 5. Impact of number of iterations on ITAE, evaluated by: (a) TD; (b) BS; and (c) ODP indicators across six datasets.

3) Overall Performance: The overall performance evaluation parameters of $\mathrm{AUC}_{(\mathrm{D},\mathrm{F})}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, $\mathrm{AUC}_{\mathrm{SNPR}}$, and $\mathrm{AUC}_{\mathrm{ODP}}$ measure the detector efficiency from a more comprehensive view. For all six datasets, at least three in four parameters generated by ITAE is the best or second best, demonstrating the superiority of ITAE among other methods. Because of the remarkable TD ability ITAE has, the $\mathrm{AUC}_{\mathrm{SNPR}}$ values of most datasets are far larger than other methods, leaded by 4581.9694 under the Urban dataset. Especially, the $\mathrm{AUC}_{\mathrm{ODP}}$ values of all datasets illustrate that the optimal and suboptimal values are all generated by ITAE, verifying the effectiveness of the proposed method. In a comprehensive view, BA-OSP and E-CEM are also admirable methods in overall performance, which exceed other detectors.

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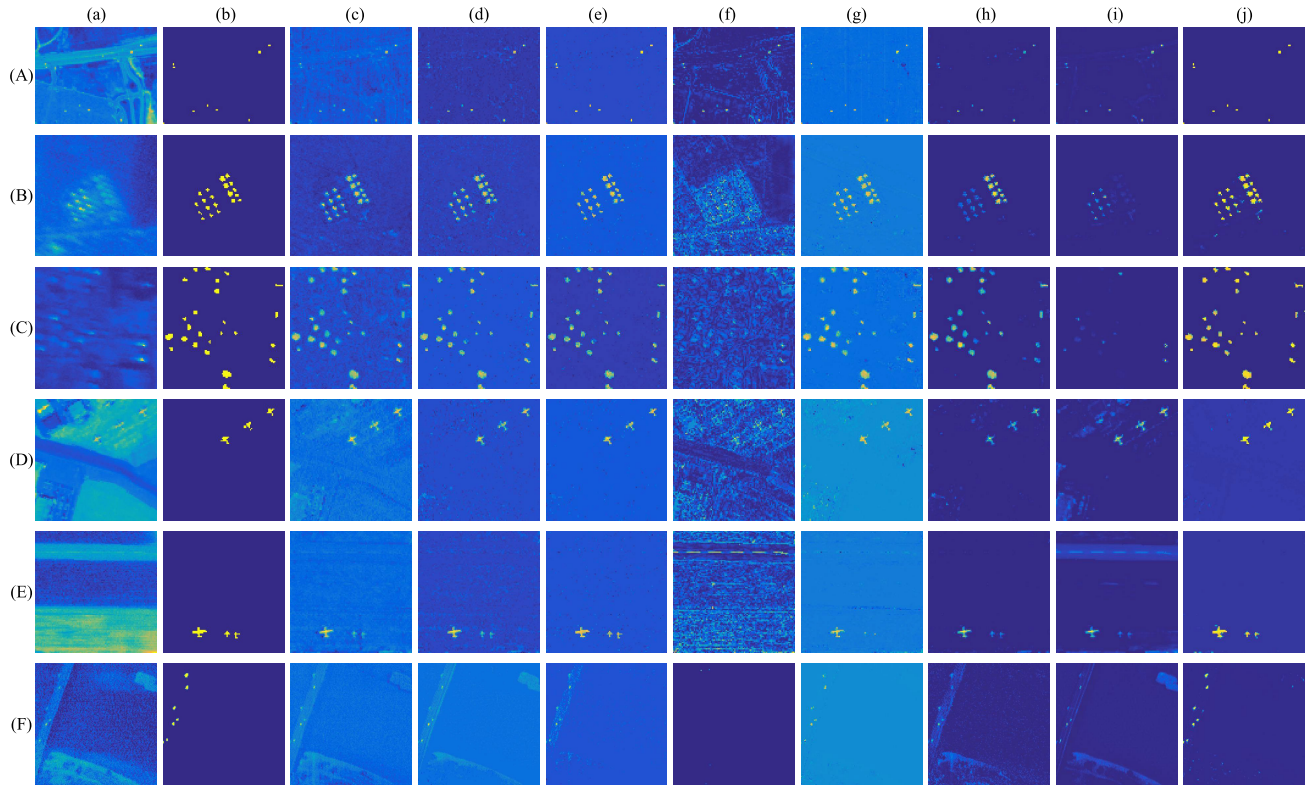


Fig. 6. Detection results comparison. (A) Urban. (B) Abu-urban-1. (C) Abu-urban-2. (D) Sandiego. (E) Abu-airport. (F) Abu-beach. (a) First-band image. (b) Ground truth. (c) CEM. (d) hCEM. (e) E-CEM. (f) DM-BDL. (g) BA-OSP. (h) BLTSC. (i) DSC. (j) ITAE.

TABLE IV
3-D ROC-DERIVED AUC RESULTS OF COMPETITIVE METHODS FOR THE URBAN DATASET

Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
CEM [29]	**1.0000**	0.6663	0.1093	1.6662	0.8907	0.5570	6.0950	1.5569
hCEM [21]	**1.0000**	0.6803	0.0478	1.6803	0.9522	0.6325	14.2199	1.6325
E-CEM [18]	**1.0000**	0.9153	0.0789	1.9153	0.9211	0.8364	11.6001	1.8364
DM-BDL [40]	0.9996	0.6426	0.0252	1.6422	0.9744	0.6174	25.5239	1.6170
BA-OSP [39]	0.9986	0.7516	0.2061	1.7503	0.7925	0.5455	3.6466	1.5442
BLTSC [54]	0.9999	0.4674	0.0008	1.4673	0.9992	0.4666	619.6162	1.4666
DSC [46]	0.9998	0.5132	0.0025	1.5130	0.9973	0.5107	206.0357	1.5105
ITAE	**1.0000**	**1.0000**	**0.0002**	**2.0000**	**0.9998**	**0.9998**	**4581.9694**	**1.9998**

Note: the bold and underlined data represent the optimal and suboptimal performance for the corresponding 3-D ROC-derived AUC results, respectively.

TABLE V
3-D ROC-DERIVED AUC RESULTS OF COMPETITIVE METHODS FOR THE ABU-URBAN-1 DATASET

Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
CEM [29]	0.9995	0.6182	0.0621	1.6177	0.9374	0.5561	9.9551	1.5556
hCEM [21]	0.9998	0.6397	0.0671	1.6394	0.9327	0.5725	9.5311	1.5723
E-CEM [18]	**1.0000**	0.7246	0.1086	1.7246	0.8914	0.6161	6.6749	1.6161
DM-BDL [40]	0.8621	0.2919	0.0822	1.1541	0.7799	0.2097	3.5496	1.0718
BA-OSP [39]	0.9997	0.7149	0.2183	1.7146	0.7814	0.4966	3.2746	1.4963
BLTSC [54]	0.9918	0.4603	0.0019	1.4521	0.9899	0.4584	**246.4342**	1.4502
DSC [46]	0.9943	0.1085	**0.0008**	1.1028	0.9935	0.1077	133.8251	1.1020
ITAE	0.9994	**0.8464**	0.0047	**1.8458**	**0.9947**	**0.8417**	180.1839	**1.8411**

Note: the bold and underlined data represent the optimal and suboptimal performance for the corresponding 3-D ROC-derived AUC results, respectively.

More intuitively, by observing the changing trends in the ITAE detection results as the number of iterations increased on each dataset, it can be found that TD and overall performance indicators are increasing, as BS indicators are decreasing iteratively. It means that the iterative framework effectively smooths and suppresses the BKG and highlights the target

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TABLE VI
3-D ROC-DERIVED AUC RESULTS OF COMPETITIVE METHODS FOR THE ABU-URBAN-2 DATASET

Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
CEM [29]	0.9986	0.6105	0.0908	1.6090	0.9077	0.5197	6.7223	1.5182
hCEM [21]	0.9960	0.6203	0.0964	1.6163	0.8996	0.5239	6.4357	1.5199
E-CEM [18]	0.9960	0.5910	0.0541	1.5870	0.9419	0.5369	10.9208	1.5329
DM-BDL [40]	0.9080	0.2100	0.0648	1.1180	0.8431	0.1451	3.2381	1.0531
BA-OSP [39]	**0.9987**	0.6853	0.1761	1.6840	0.8226	0.5093	3.8923	1.5079
BLTSC [54]	0.9973	0.5556	0.0065	1.5530	0.9909	0.5492	86.1230	1.5465
DSC [46]	0.8993	0.0456	**0.0006**	0.9448	0.8987	0.0450	79.5596	0.9443
ITAE	0.9969	**0.7753**	0.0044	**1.7722**	**0.9926**	**0.7709**	**176.7918**	**1.7678**

Note: the bold and underlined data represent the optimal and suboptimal performance for the corresponding 3-D ROC-derived AUC results, respectively.

TABLE VII
3-D ROC-DERIVED AUC RESULTS OF COMPETITIVE METHODS FOR THE SANDIEGO DATASET

Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
CEM [29]	0.9972	0.6988	0.1879	1.6960	0.8094	0.5109	3.7192	1.5081
hCEM [21]	0.9994	0.6408	0.0876	1.6401	0.9117	0.5531	7.3125	1.5525
E-CEM [18]	**1.0000**	0.7002	0.1074	1.7002	0.8925	0.5928	6.5172	1.5928
DM-BDL [40]	0.9332	0.4811	0.0947	1.4143	0.8385	0.3864	5.0800	1.3196
BA-OSP [39]	0.9953	0.7302	0.2911	1.7255	0.7041	0.4390	2.5080	1.4343
BLTSC [54]	0.9953	0.4144	**0.0024**	1.4096	**0.9928**	0.4119	**169.4924**	1.4072
DSC [46]	0.9949	0.5836	0.0083	1.5785	0.9866	0.5754	70.6663	1.5703
ITAE	0.9992	**0.8938**	0.0094	**1.8930**	0.9898	**0.8844**	94.8367	**1.8836**

Note: the bold and underlined data represent the optimal and suboptimal performance for the corresponding 3-D ROC-derived AUC results, respectively.

TABLE VIII
3-D ROC-DERIVED AUC RESULTS OF COMPETITIVE METHODS FOR THE ABU-AIRPORT DATASET

Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
CEM [29]	0.9994	0.5829	0.1561	1.5823	0.8433	0.4268	3.7339	1.4262
hCEM [21]	0.9998	0.5860	0.0787	1.5858	0.9211	0.5074	7.4485	1.5071
E-CEM [18]	**1.0000**	0.7471	0.0951	1.7471	0.9049	0.6520	7.8566	1.6520
DM-BDL [40]	0.6381	0.1603	0.1212	0.7984	0.5169	0.0391	1.3226	0.6772
BA-OSP [39]	0.9367	0.5110	0.2226	1.4478	0.7141	0.2884	2.2957	1.2251
BLTSC [54]	0.9993	0.3832	**0.0011**	1.3825	**0.9982**	0.3821	**361.7336**	1.3814
DSC [46]	0.9772	0.3117	0.0113	1.2889	0.9659	0.3004	27.5743	1.2776
ITAE	0.9999	**0.8694**	0.0492	**1.8693**	0.9506	**0.8202**	17.6567	**1.8200**

Note: the bold and underlined data represent the optimal and suboptimal performance for the corresponding 3-D ROC-derived AUC results, respectively.

TABLE IX
3-D ROC-DERIVED AUC RESULTS OF COMPETITIVE METHODS FOR THE ABU-BEACH DATASET

Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
CEM [29]	0.9419	0.4132	0.1704	1.3551	0.7715	0.2427	2.4241	1.1847
hCEM [21]	0.8366	0.4103	0.1869	1.2469	0.6497	0.2234	2.1950	1.0600
E-CEM [18]	0.9548	0.4045	0.0996	1.3593	0.8552	0.3049	4.0618	1.2597
DM-BDL [40]	0.8020	0.0014	**0.0006**	0.8034	0.8015	0.0008	2.3720	0.8028
BA-OSP [39]	0.9473	0.5375	0.2843	1.4848	0.6630	0.2532	1.8906	1.2005
BLTSC [54]	0.9589	0.2421	0.0148	1.2010	0.9441	0.2273	16.3487	1.1862
DSC [46]	0.6047	0.1534	0.0127	0.7581	0.5920	0.1408	12.1057	0.7455
ITAE	**0.9800**	**0.6537**	0.0135	**1.6338**	**0.9666**	**0.6403**	**48.5605**	**1.6203**

Note: the bold and underlined data represent the optimal and suboptimal performance for the corresponding 3-D ROC-derived AUC results, respectively.

layer by layer since the quality of the image is improved gradually.

Figs. 7–12 establish the 3-D ROC curves along with the corresponding 2-D ROC curves of the competitive methods for six datasets, where the logarithmic scale is used on the x-axis of the (P_D, P_F) curves for the convenience of visual comparison. We analyze the detection performance by the following criterions.

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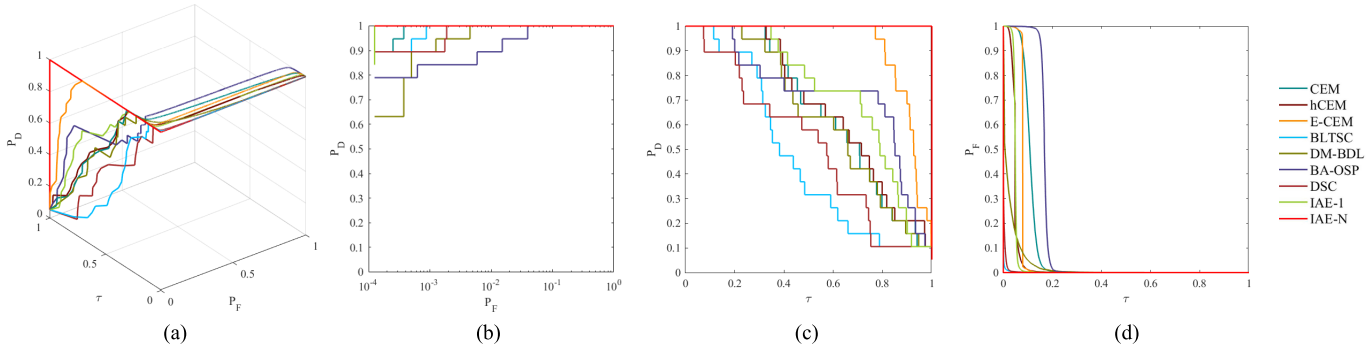


Fig. 7. 3-D ROC curves along with the corresponding 2-D ROC curves of the competitive methods for the Urban-1 dataset. (a) 3-D ROC curves. (b) 2-D ROC curves of (P_D, P_F). (c) 2-D ROC curves of (P_D, τ). (d) 2-D ROC curves of (P_F, τ).

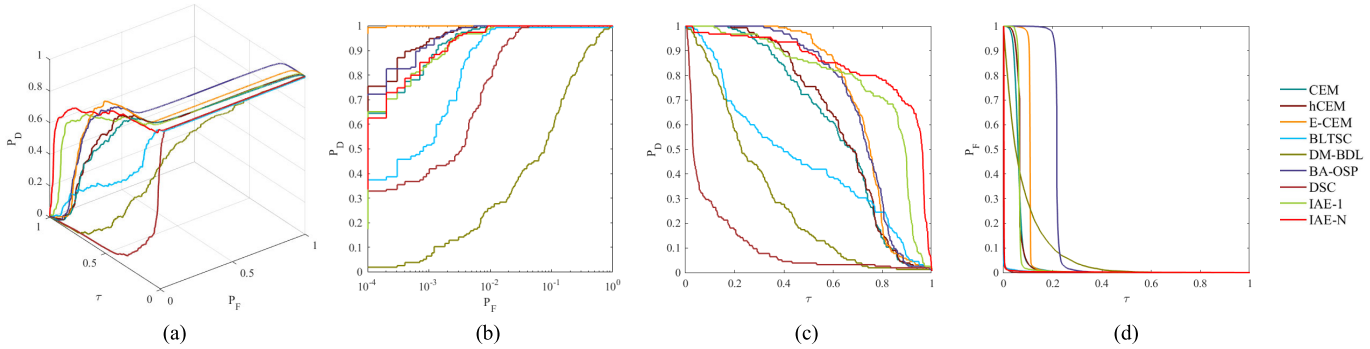


Fig. 8. 3-D ROC curves along with the corresponding 2-D ROC curves of the competitive methods for the Abu-urban-1 dataset. (a) 3-D ROC curves. (b) 2-D ROC curves of (P_D, P_F). (c) 2-D ROC curves of (P_D, τ). (d) 2-D ROC curves of (P_F, τ).

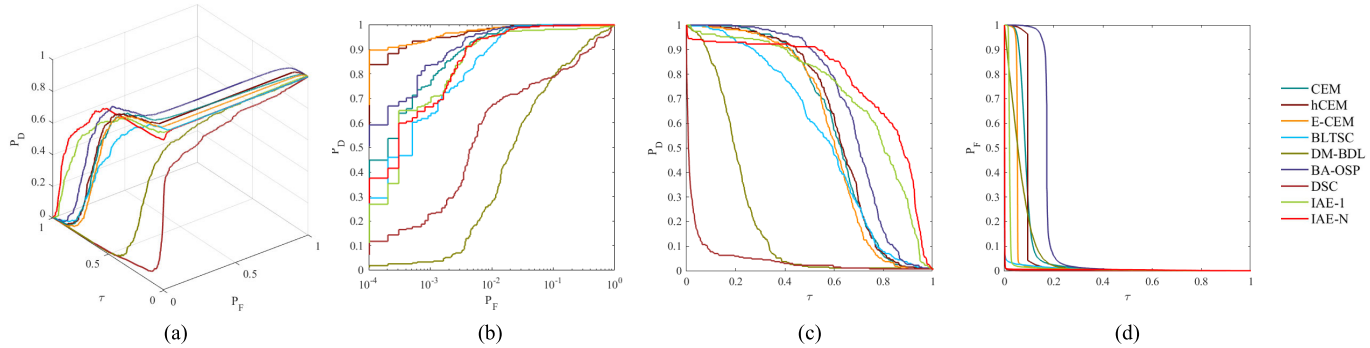


Fig. 9. 3-D ROC curves along with the corresponding 2-D ROC curves of the competitive methods for the Abu-urban-2 dataset. (a) 3-D ROC curves. (b) 2-D ROC curves of (P_D, P_F). (c) 2-D ROC curves of (P_D, τ). (d) 2-D ROC curves of (P_F, τ).

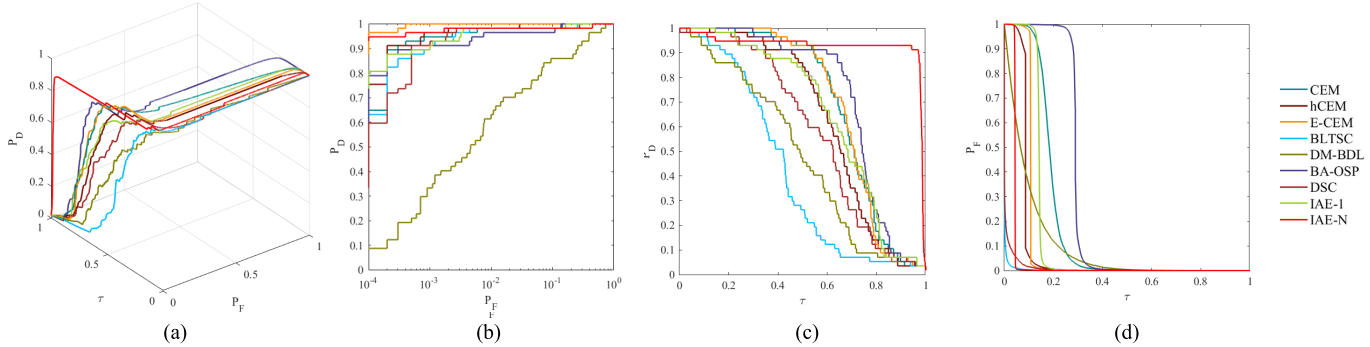


Fig. 10. 3-D ROC curves along with the corresponding 2-D ROC curves of the competitive methods for the Sandiego dataset. (a) 3-D ROC curves. (b) 2-D ROC curves of (P_D, P_F). (c) 2-D ROC curves of (P_D, τ). (d) 2-D ROC curves of (P_F, τ).

1) For curves (P_D, P_F) and (P_D, τ), a larger area under the curve indicates better detection ability, and for curve (P_F, τ), we want the area be smaller for better BS.

2) For curve (P_D, P_F), closer distance from the upper left corner in the starting point and faster arrival on the $P_D = 1$ point mean better effectiveness.

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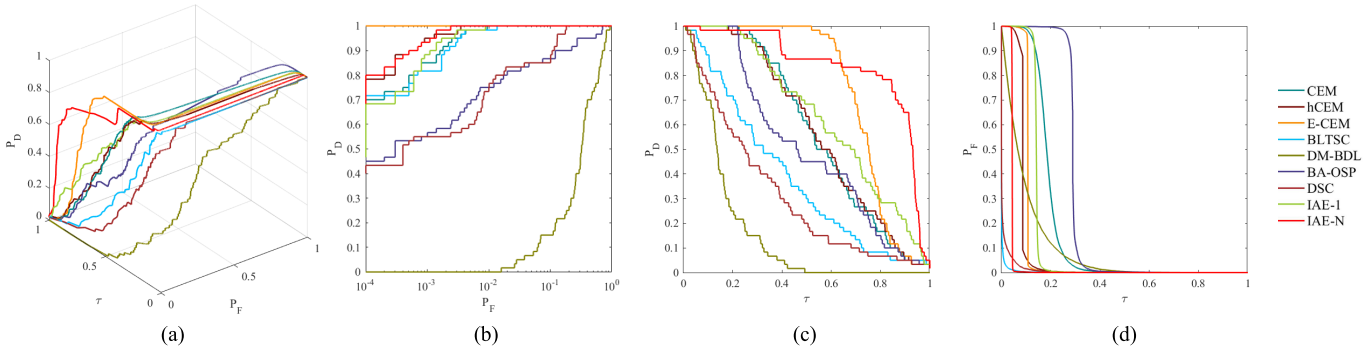


Fig. 11. 3-D ROC curves along with the corresponding 2-D ROC curves of the competitive methods for the Abu-airport dataset. (a) 3-D ROC curves. (b) 2-D ROC curves of (P_D, P_F). (c) 2-D ROC curves of (P_D, τ). (d) 2-D ROC curves of (P_F, τ).

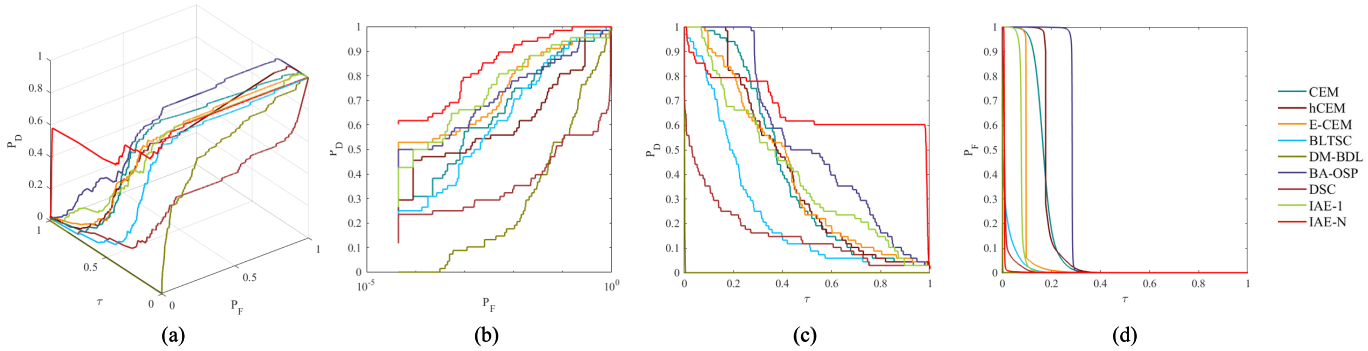


Fig. 12. 3-D ROC curves along with the corresponding 2-D ROC curves of the competitive methods for the Abu-beach dataset. (a) 3-D ROC curves. (b) 2-D ROC curves of (P_D, P_F). (c) 2-D ROC curves of (P_D, τ). (d) 2-D ROC curves of (P_F, τ).

3) For curve (P_F, τ), the steeper the downward slope of the curve, the better the BS of the detector.

Based on the above evaluation criteria, we found that the proposed ITAE method performs best in TD, BS, and overall performance, which is consistent with the analysis results of the ROC indexes and target maps. Additionally, among six datasets, we found that in the complex scenes like airport and beach, the BS ability of ITAE is not significant as in other simpler scenes, while the TD ability stays stably distinguished.

E. Noisy Scene Analysis

To verify the detector's robustness to noisy scenes, we conducted a series of experiments on the six datasets with added Gaussian noise. To avoid the unfairness brought by the datasets' varying average reflectance, the variance of the Gaussian noise is obtained by a certain SNR (dB). The concrete calculation is as follows:

$$\sigma^2 = \frac{E\left[X^2\right]}{10^{\frac{\mathrm{SNR}}{10}}} \tag{22}$$

where σ refers to the variance of the Gaussian noise, and $E[X^2]$ refers to the expectation value of the square of the HSI.

In this article, we added two different degrees of Gaussian noise, SNR = 5 dB and SNR = 10 dB, in the original HSIs. The first band image of the noisy HSI and the detection result of the ITAE is shown in Fig. 13. It can be found that under two degrees of noise, the ITAE methods are capable of detecting the interested targets, and the targets are more discriminative in the SNR = 10 dB scenario than those in the SNR = 5 dB scenario. Notably, the BKG in the Sandiego datasets under SNR = 5 dB noise is relatively weakly suppressed compared to other datasets, considering the airport scene is more complex and the spectral similarity between man-made objects is high. Basically, the TD and BS ability that ITAE shows under noisy circumstances are promising.

F. Ablation Analysis

To demonstrate the effectiveness of the proposed modules, we take the Urban dataset as an example and evaluate the detection performance when ITAE annihilates these modules gradually: 1) VD constraint of target-biased AE; 2) TKL-weighted loss of target-biased AE; and 3) iterative structure with CEM-add image renew. Especially, for fairer comparison, we set the number of iterations in this section as 4 to control the variables. The detection results of 3-D-derived AUC values are listed in Table X.

1) Effectiveness Analysis of VD Constraint: According to Section III-A1, for noise and interference annihilation, the hidden dimension of the target-biased AE is constraint with VD. To prove that this constraint helps to improve TD and BS, we conduct the ITAE with and without VD constraint severally. Notably, when the network is not constrained with hidden dimensionality, we set the dimensionality of latent code as an empirical value, 30. The results in Table X show that all the indicators become worse after removing the constraint.

2) Effectiveness Analysis of TKL-Weighted Loss: According to Section III-A2, before reconstruction, the TKL map of the HSI is calculated as a weighting parameter consisting a part of

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TABLE X
3-D ROC-DERIVED AUC RESULTS OF ABLATION ANALYSIS FOR THE URBAN DATASET

Iterative Structure	TKL-weighted Loss	VD Constraint	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}
✓	✓	✓	**1.0000**	**1.0000**	**0.0002**	**2.0000**	**0.9998**	**0.9998**	**4581.9694**	**1.9998**
✓	✓		**1.0000**	0.9985	0.0011	1.9985	0.9989	0.9974	900.1502	1.9974
✓			**1.0000**	0.9984	0.0012	1.9984	0.9988	0.9972	817.4718	1.9972
			1.0000	0.8037	0.0497	1.8037	0.9502	0.7540	16.1657	1.7540

Note: the bold and underlined data represent the optimal and suboptimal performance for the corresponding 3-D ROC-derived AUC results, respectively.

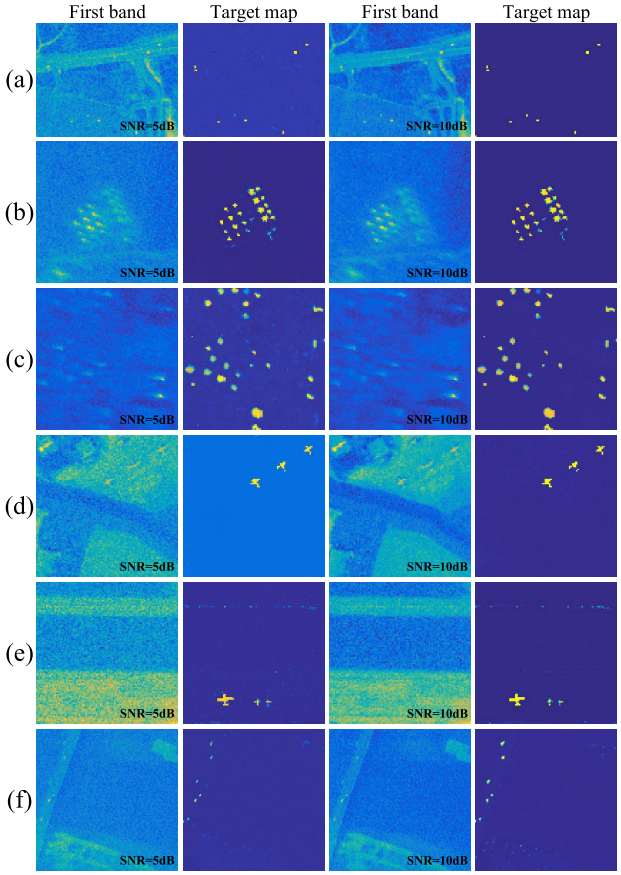


Fig. 13. Detection results on noisy scenes with SNR = 5 and SNR = 10. (a) Urban. (b) Abu-urban-1. (c) Abu-urban-2. (d) Sandiego. (e) Abu-airport. (f) Abu-beach.

the loss function in the following target-driven AE. It is meant to guide the reconstruction network pay more attention on the reconstruction of pixels, which are more likely to be the target and infer more details of the target and smooth the BKG in the inference process. After removing the TKL calculation step, we set the loss function of AE as the typical MSE function of the images before and after reconstruction. As shown in Table X, the detection performance further decreased without TKL-weighted loss, proving its efficacy.

3) Effectiveness Analysis of Iterative Structure: According to Section III-B, we design an iterative structure for image quality improvement gradually, providing better detection results layer by layer. In Section IV-D, from the trend of 3-D ROC-derived AUC values that ITAE generated as the number of iterations increases, the efficiency of iterative structure can be proved. To further validate this, Table X also shows the importance of iterative structure more concretely.

V. CONCLUSION

In this article, a general framework for HTD is presented, which introduces an iterative system into an AE-based network collaboratively, called ITAE. Iteratively, the target component is enhanced supported by the posterior knowledge. The iterative core, called TAE, implements a prioritized reconstruction driven by target characteristics and achieves data quality improvement. Incorporating the classic HTD and DL methods, ITAE greatly improves TD and has strong robustness to noise. There are several novelties.

1) The iterative loop is included into an AE-based network and forms a closed-end system, where the diversity between the target and other components is improved step by step until terminated by an automatic stopping rule.
2) ITAE strengthens the utilization of posterior knowledge by a feedback branch, enabling further exploration of the target characteristics. Iteratively, the targets get enhanced supported by both prior and posterior knowledge.
3) Unlike typical AE that implements an identity mapping, the core of ITAE, named TAE, can adaptively assign the diverse priorities pixelwise during reconstruction. This results in a better learning of the key components, targets, while neglecting the restoration of BKG, noise, and interference details, and contributes to better TD.
4) Without abundant layers and connections, TAE uses a simple structure where the only structural variant, the hidden dimensionality of latent code, can also be adaptively determined. By constraining the number of neurons by the estimated VD, the noise and interference components are selectively ignored in the reconstruction process, denoting a strong robustness to noise and interference.

To prove the efficiency of ITAE, a series of experiments is conducted compared to other seven state-of-the-art methods. Results show the excellent performance of ITAE, especially in TD and prove its robustness to noise of different degrees. To demonstrate the effectiveness of proposed novelties, the ablation study was conducted. And the sensitivity analysis was provided to validate the robustness of our method against parameter settings and dataset uncertainties. In the future work, we will investigate deeper into the potential of iterative framework, finding more effective cores and optimizing the network structure.

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